

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)**(TRIMESTER –II) END TERM EXAMINATION, JANUARY -2024****Quantitative Techniques****[Time: 3 Hours]****[Maximum Marks: 60]****Following are the course outcomes of the subject: - Quantitative Techniques**

CO	Course Outcome (CO)
CO1	Understand the fundamentals of quantitative techniques and its interdisciplinary role in decision making
CO2	Identify the right models for solving the particular problem
CO3	Apply the models for decision making and forecasting.

(Note- Faculty can add the more CO's as per their subject.)**SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)**

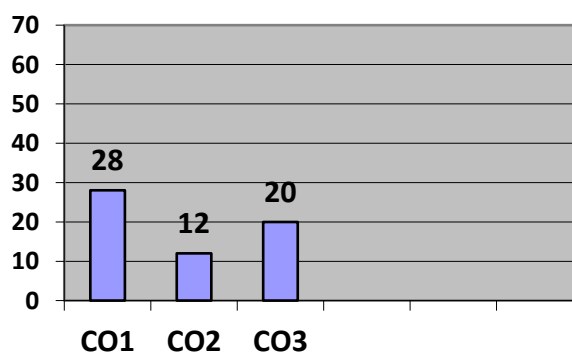
Q.1	Marks	CO
(A)	4	CO – 1
(B)	4	CO – 1
(C)	4	CO – 1
(D)	4	CO – 1
(E)	4	CO – 1

SECTION-B**Attempt all questions out of the following: (Long Answer Type Questions)**

Q. No	Marks	CO
2 (A)	6	CO – 2
2 (A)	6	CO – 2
2 (B)	6	CO – 2
2 (A)	6	CO – 2
3 (A)	6	CO – 3
3 (A)	6	CO – 3
3 (B)	6	CO – 3
3 (B)	6	CO – 3

SECTION-C**Attempt all parts of the following: (Caselet / Numerical)**

Q.4	Marks	CO
4 (A)	8	CO – 3
4 (B)	8	CO – 1

Course Outcome Wise Marks Distribution

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GNIOT Institute of Management Studies – PGDM (Batch 2023-25)**(TRIMESTER – II) END TERM EXAMINATION, JANUARY - 2024****Quantitative Techniques****[Time: 3 Hours]****[Maximum Marks: 60]****INSTRUCTIONS:**

- (i) **There are THREE sections in this question paper.**
(ii) **Attempt questions from each section as per the directions.**
(iii) **Make suitable assumptions wherever necessary.**

TVS motors is one of India's largest two-wheeler manufacturing firms. The production plants of TVS are located at four places in India. TVS also invest heavily on research and development. The domain of research comprehensively covers product development, product improvement, operations management, market capitalization, competitor's analysis etc.

The company is willing to launch a new motorbike in premium segment in year 2024. The company wishes to understand the customers' expectation from the new launch.

SECTION-A**(Short Answer Type Questions, do not exceed 200 words)****Q.1. Attempt all parts of the following:****(5×4=20)**

- (a) Shall company go for population survey or survey? Give a brief reasoning for your answer.
(b) If the company has decided to go for a sampling survey, which sampling method you think TVS shall opt for understanding the customers' expectation and why?
(c) TVS is also willing to estimate the expected price by the customers for the vehicle. Which estimate would be better point estimation or interval estimation and why?
(d) If TVS is willing to identify the independence of variables, which of the statistical techniques they will be using on the collected data?
(e) By observing the market TVS research team has projected that the acceptance of this newly launch motorbike would be between 7 – 9 on a acceptance level 10 point scale. Where 10 means highest level of acceptance. What would be null and alternate hypothesis in this case.

SECTION-B**(Long Answer Type Questions)**

One of the key measurements during market research is to identify the level of satisfaction from milage with respect to the age group of the potential consumers. The customers are divided into three age groups and the question is asked, how important the mileage is if they are buying a sports bike? Following is the data

Age Group	Importance of milage from a sports bike	
	Significantly Important	Not Important
18 – 25 years	23	38
25 – 35 years	40	42
Above 35 years	50	27

Q.2. Attempt all questions of the following**(4×6=24)**

- (a) Find the chance that a randomly selected person is either of 25 – 35 years or say the milage is not important. Given that the person is of 18.- 25 years what is the probability that the randomly selected person quotes that the milage is significantly important.

OR

- (a) Find the chance that a randomly selected person is of 25 – 35 years and claims that milage is not important. Given that the person is quoting milage not important, what is the probability that person is above 35 years.

After conducting the market survey the TVS motor company launched the motor bike. The bike was made available on the dealer showrooms across the country.

While conducting the market survey, it was found that the people in the age group 25 – 35 years have a 80% buying potential for the motorbike. If 4 persons of age group 25 – 35 come to visit the show room in one hour.

- (b) Calculate the probability that 2 or more customers end up buying the motorbike?

OR

- (b) If the dealer showroom receives 0.9 inquiry calls per minute. Calculate the probability that in an interval of four minutes there will be at-most three incoming calls ($e^{-3.6}=0.027$)

Q.3. (a) The average sales obtained weekly by showrooms is 65 units with a standard deviation of 8. The top 10% showroom are to receive the incentives. What is the minimum sales a showroom shall achieve to get the incentives.

OR

- (a) Further the data shows that the monthly sales of such motorbikes of competitors was 3000 bikes with a standard deviation of 120 bikes. A sample of 40 TVS showrooms shown a monthly sales of 2940 bikes. At 5% level of significance, can it be claimed that the sales of TVS bikes is significantly lower than the competitors.

TVS motors also conducted a post sales satisfaction survey for their newly launched vehicle. The satisfaction measurement scale kept as a 100 point score, where 100 means the highest level of satisfaction.

- (b) A sample of 25 customers is taken and it is found that their satisfaction level score is found 85 with a standard deviation of 9. While it was claimed that the satisfaction score will remain above 88. At 5% level of significance, can it be claimed that the sample result is significantly different from the claim.

OR

- (b) The data is also collected on satisfaction of customers from milage w.r.t. their income group. Following table presents the data

Income (per annum)	Level of satisfaction	
	Satisfied	Not-satisfied
Less than Rs. 4,50,000	26	34
Between Rs. 4,50,000 – Rs. 8,00,000	40	37
More than Rs. 8,00,000	32	24

Can it be claimed at 5% level of significance that income group is independent of level satisfaction.

SECTION-C

TVS motors went a step ahead in conducting their research. They have contracted 3 different riders and given them two different models of competitors and TVS model to ride 5000 kms across India alternatively. Their comprehensive level of satisfaction from the performance of vehicle is collected. Following table shows the results;

	TVS	Competitor 1	Competitor 2
Rider 1	7	6	7
Rider 2	6	7	8
Rider 3	7	6	6

Q.4. Attempt all parts of the following:

(2×8=16)

- 4(a)** at 5% level of significance can it be claimed that, the variation in the satisfaction scores is purely a matter of chance. State and formulate null and alternate hypothesis and test it.
- 4(b)** State, the idea of estimation in detail. How type I and type II errors come in picture while testing the hypothesis. How do you differentiate null and alternate hypothesis. State the steps of hypothesis testing.